

A Minimalist Social Robot Platform for Promoting Positive Behavior Change Among Children

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ABSTRACT

We present the design of a minimalist interactive social robot named Haksh-E which integrates audio-visual perception with artificial intelligence to nurture positive behaviors in children. We co-designed the robot's embodiment with children acting as design informants. The robot features expressive interactions, real time conversation and human action detection capabilities. Haksh-E can assist children in learning new behaviors as a tutor, play a supportive role in health interventions and also pose as an objective assessor of various indicators of health and hygiene.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *Empirical studies in HCI*; *Interaction design process and methods*.

KEYWORDS

Design, Robotics, Education, Interaction

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1 INTRODUCTION

With recent advancements in AI and robotics, physically embodied robots can act as emotionally-intelligent social actors. This allows for research into pro-social computing where non-human, embodied models, such as social robots are used for promoting positive

behavior among children for their overall health and well-being [Peter et al. 2021]. In this talk, we present a scalable social robot platform that we co-designed with children to promote hand hygiene (refer to Figure 1). The platform consists of a deep learning based hand washing action recognition system (HAR) and a perception system that supports the conversational and emotional communication capabilities of the robot. For our behavior change case study, we created a platform based on minimalistic design principles to allow for large-scale deployments at affordable costs and low design complexities. We utilized form factors that could leverage fabrication technologies such as 3D printing for rapid prototyping and off-the-shelf components for assembling the physical prototype. In this talk, we discuss the robot's embodiment design and interaction capabilities.

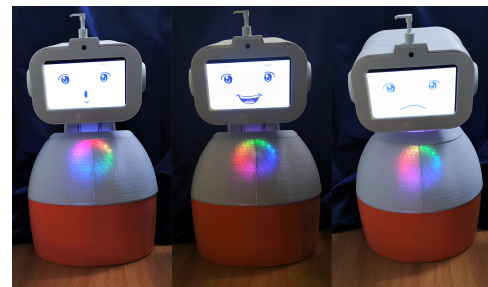


Figure 1: Facial expressions displayed by *Haksh-E* version 0.5

2 EMBODIMENT DESIGN

We adopted a participatory approach for co-designing the robot with children where they played the role of informants. We chose forty children from various socioeconomic backgrounds across India. To explore children's mental representations of a robot which promotes hand washing behavior, we provided them with a variety of pre-cut geometrical shapes made from paper and asked them to create their own models of the robot (refer to Figure 2). We then asked the children to give their feedback on eight a priori conceptual designs of minimalist caricatured social robots. We took inspiration from animated feature films to create our conceptual

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designs and modeled the robot's morphology on practical everyday objects that could suggest hand washing to the children such as a soap dispenser. Our conceptual robot designs projected a degree of "humanness" for the children to feel comfortable with and yet had a degree of "robotness" so as not to create false expectations about the robot's capabilities.

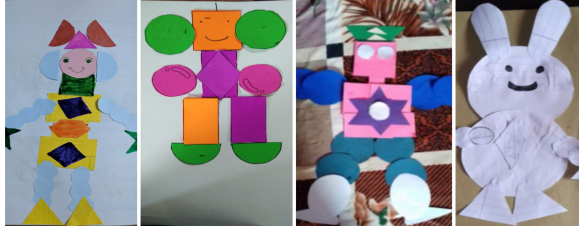


Figure 2: Robot designs made by the participants

Our study results showed that the children unequivocally liked and trusted the anthropomorphised caricatured designs of everyday objects for the robot's morphology. Also, the children had high expectations on emotional intelligence, interactions and social roles of the robot even though it was being designed for a specific context to promote hand hygiene. Based on the descriptions the children provided for their hand washing buddy and looking into social robot design principles from Child-Robot Interaction literature, we designed the final prototype for Haksh-E (refer to Figure 3 and Figure 4).

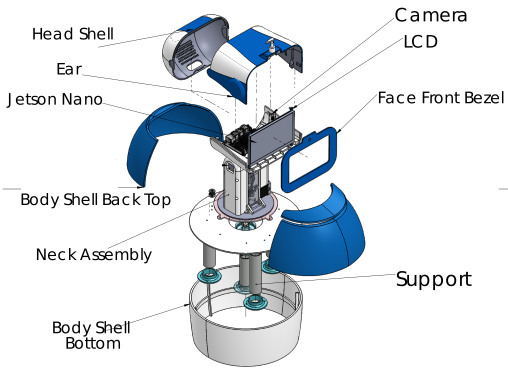


Figure 3: Exploded view of the robot embodiment

3 INTERACTION CAPABILITIES

Affect display. - The robot is built to act as an agent with a high degree of autonomy during its interaction with a child. To design an affective behavioral model for the robot, it must empathise with the child and respond using ideal combination of verbal and non-verbal responses. The affective behavioral model is based on an action-response sequence whose decisions are governed using an AI based agent. This improves the quality of child-robot interaction and makes the interaction more life-like, while the robot learns to adapt with the environment.

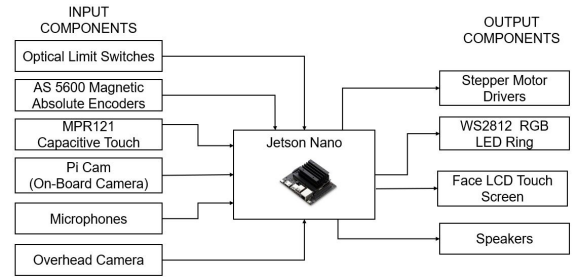


Figure 4: Prototype Design - A high-level block diagram of hardware design

The animated facial expressions of Haksh-E are simulated on the LCD screen present on its head. The face of Haksh-E has eyebrows and human-like, large eyes with irises so that it is more expressive and accurate with its gaze. Based on cognitive science research on how colours are judged affectively [Briki and Hue 2015], we attached an RGB LED ring within the body of the robot to glow in different colours, expressing the current emotional state of the robot.

Action Detection. - The robot features an AI aided hand washing action recognition (HAR) system that can detect the quality of hand washing based on the hand washing steps prescribed by the World Health Organization (WHO). Real time feedback is provided to the children, either to correct them if their procedure of hand washing is wrong, or to appreciate and motivate them to maintain the quality of hand washing, if they performed hand washing in the right way. The HAR system on the overhead camera is integrated with vision system of the robot platform and it is trained on an augmented dataset using a modified Convolutional Neural Network (CNN) architecture with a self-gated attention module and a residual architecture as the backbone.

Conversational AI. - The robot is capable of having conversations around hand hygiene and also supports limited non-linear conversations such as telling jokes and enquiring about the well-being of the child in a gender neutral voice. It can also talk about the 6 steps of hand washing as prescribed by WHO. The conversational AI is built using a combination of RASA - an open-Source machine learning framework and Google's Automatic Speech Recognition and Text-To-Speech engines.

Thus, by combining a HAR system, a conversational AI and an emotionally aware perception system, Haksh-E has the capability to act as a social agent that can interact with children, build relationships with them and help them develop positive habits over the long term.

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